

What's Tick-Tock

Tick-Tock is a model developed by Ashwani Gupta of Jones Farm 5 (Hillsboro, Oregon) which has been adopted by Intel Corporation since 2007 to follow every microarchitectural change with a die-shrink of the process technology. Every 'tick' is a shrinking of the process technology of the previous microarchitecture and every 'tock' is a new microarchitecture.

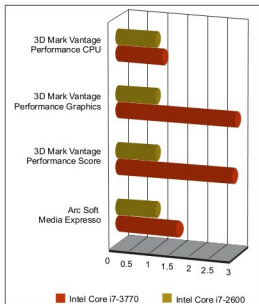


Fig. 3: Performance of Intel Core i7-2600 vs Intel Core i7-3770 (Disclaimer: This representation utilises leaked data to populate the chart)

cessor can be overclocked for higher performance—useful for gaming enthusiasts and other power users. 'M' signifies that it is a processor for use on a mobile platform.

Sandy Bridge is, in fact, a major tock. The architecture underwent a major platform repartitioning with a lot of things being integrated on the die. For instance, the clock generator and the integrated graphics processor now share space with the CPU. All this integration results in extremely low power consumption. The on-die graphics processor called Intel HD Graphics delivers a below par performance, but improved power efficiency through the power-gated GPU. This made Sandy Bridge the best option for portable PCs.

Sandy Bridge was released in January 2011.

Ivy Bridge

Intel's 22nm die-shrink of Sandy Bridge is codenamed Ivy Bridge. Apart from the die-shrink, Intel has utilised a non-planar architecture called tri-gate for the transistor in Ivy Bridge. Marketed as 3D

transistors, this technology is claimed to allow the processors to cut power consumption by half and perform 37 per cent faster. Performance wise, the CPU is expected to outperform Sandy Bridge by a whopping 20 per cent.

From Fig. 3, you can see that Intel HD Graphics processor in Ivy Bridge significantly outperforms those in Sandy Bridge processors.

Although Intel factories had to be upgraded to be able to manufacture these tri-gate Ivy Bridge CPUs, the market prices remain roughly the same as a finished wafer costs only about 3 per cent more.

Ivy Bridge also overcomes the shortcomings of the previous line of processors by improving on the graphics processing capabilities. The Intel HD Graphics in Ivy Bridge supports DirectX 11, OpenGL 3.1 and OpenCL 1.1, while also outnumbering the executing units by four to reach a maximum of 16. Intel claims the graphics performance to be 30 per cent better than the previous-generation Intel HD Graphics. Intel HD Graphics in Ivy Bridge comes in two variants: Intel HD 2500 for low-end performance and Intel HD 4000 for better performance.

Mobile processors also support DDR3L RAM. This is a low-voltage RAM which operates at 1.35V compared to the standard 1.5V. This results in a 15 per cent reduction in power consumption by the RAM itself.

Ivy Bridge was released in April this year. Upgrading to Ivy Bridge would be a straightforward affair since both Ivy and Sandy use the same LGA 1155 socket.

Haswell: The next tock

Haswell is the codename for Intel's next microarchitecture which forms the next 'tock' in Intel's roadmap. Similar to Ivy Bridge, Haswell is based on the 22nm manufacturing process using tri-gate technology. This processor is not backward-compatible as it uses the LGA1150 socket. Ivy Bridge and Sandy Bridge use the LGA 1155 socket. So to get a Haswell mi-

Processor Architectures Over The Years

Architecture	Year	Processors that use it	Comments
NetBurst	2000	Pentium 4, Pentium D, Xeon	Came in 180nm, 130nm and 90nm versions over the years
Core	2006	Core 2, Xeon	Dual-core based on P6 microarchitecture, 65nm process
Penryn	2007	Core 2, Xeon	45nm die-shrink of Core
Nehalem	2008	Core i3, Core i5, Core i7	The first versions of the Core-i series of processors at 45 nm
Westmere	2010	Core i3, Core i5, Core i7	Die-shrink of Nehalem at 32 nm with the same branding
Sandy Bridge	2011	Core i3, Core i5, Core i7	New microarchitecture with the same Core-i branding at 32 nm
Ivy Bridge	2012	Core i3, Core i5, Core i7	Die-shrink of Sandy Bridge at 22 nm, still with the same branding
Haswell	2013	NA	Future Intel microarchitecture expected next year

Note: Blue = Tick; Yellow = Tock